CAMPAIGN RESULTS

Exact Paper Model Bank Campaign Results Report

RCIM Exact-Paper Campaign Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the exact-paper model-bank campaign prepared in:

- `doc/reports/campaign_plans/2026-04-10-17-04-41_exact_paper_model_bank_campaign_plan_report.md`

  The campaign executed `4` repository-owned validation runs through the dedicated exact-paper launcher:
- completed runs: `4`
- failed runs: `0`
- execution window: `2026-04-10T18:44:39.0700428+02:00` to `2026-04-10T19:14:18.3733939+02:00`
- campaign artifact root:

  `output/training_campaigns/exact_paper_model_bank_campaign_2026_04_10_17_04_41/` Unlike the harmonic-wise `Track 1` campaign, this package does not optimize the full TE-curve offline benchmark directly. Its purpose is to validate the stricter recovered-paper branch that uses:
- recovered `rpm`, `deg`, `tor` inputs;
- exact `ampl_k` and `phase_k` targets for harmonics

  `0, 1, 3, 39, 40, 78, 81, 156, 162, 240`;
- the full recovered family bank;
- one ONNX export artifact per family and target.

Objective And Outcome

The campaign had four concrete questions

1. can the full bank finish in diagnostic mode without ONNX export crashes?
2. can the same bank finish in strict mode after the `HGBM` converter fix?
3. does the isolated `SVR` branch still need surrogate safeguards, and does it

   now serialize cleanly?
4. can the non-`SVR` branch export cleanly as a reference bank?

   Outcome:

- the full exact-paper bank now completes in both diagnostic `continue` mode

  and strict `reference` mode;
- the `HGBM` ONNX export blocker is resolved under the current local stack;
- the isolated `SVR` branch exports all `20` targets successfully, with `5`

  constant-surrogate replacements still required for degenerate targets;
- the non-`SVR` branch exports its full `180`-file surface cleanly;
- the canonical winner is the strict full-bank reference run

  `exact_full_bank_strict_reference`.

Ranking Policy

This campaign is a stabilization campaign, not a generic hyperparameter sweep. The ranking therefore follows this explicit serialized policy:

- primary metric: `winning_mean_component_mape_percent`
- first tie breaker: `winning_mean_component_mae`
- second tie breaker: prefer `strict` export mode over `continue`
- third tie breaker: prefer broader enabled-family coverage
- fourth tie breaker: `winning_mean_component_rmse`
- fifth tie breaker: lexical `run_name`

  That policy intentionally promotes the cleanest faithful reference when the diagnostic and strict runs produce equivalent validation metrics.

Campaign Ranking

Ranked Completed Runs

Rank	Config	Family Scope	Export Mode	Winner	Mean Component MAPE [%]
1	`exact_full_bank_strict_reference`	`all 10` recovered families	`strict`	`RF`	`18.369`
2	`exact_full_bank_diagnostic_continue`	`all 10` recovered families	`continue`	`RF`	`18.369`
3	`exact_non_svr_export_reference`	`9` non- `SVR` families	`strict`	`RF`	`18.369`
4	`exact_svr_export_diagnostic`	`SVR` only	`continue`	`SVR`	`103.265`

Config	Mean Component MAE	ONNX Exported	Surrogates	Failed Exports
`exact_full_bank_strict_reference`	`0.056284`	`200`	`5`	`0`
`exact_full_bank_diagnostic_continue`	`0.056284`	`200`	`5`	`0`
`exact_non_svr_export_reference`	`0.056284`	`180`	`0`	`0`
`exact_svr_export_diagnostic`	`0.138346`	`20`	`5`	`0`

Campaign Winner

The explicit campaign winner is

- `exact_full_bank_strict_reference`

  Its result was:
- winner family: `RF`
- winner family estimator: `RandomForestRegressor`
- enabled families: `SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM`
- winner mean component MAPE: `18.369%`
- winner mean component MAE: `0.056284`
- winner mean component RMSE: `0.144839`
- ONNX exported files: `200`
- failed exports: `0`
- surrogate exports: `5`

  This run won because it matched the best achieved component metric while also proving that the full exact-paper bank now survives strict export mode without fallback continuation behavior.

Top Family Ranking Inside The Winning Run

Rank	Family	Mean Component MAPE [%]	Mean Component MAE	Mean Component RMSE
1	RF	18.369	0.056284	0.144839
2	HGBM	20.586	0.079797	0.155340
3	ERT	21.017	0.054424	0.139747
4	GBM	24.343	0.062593	0.152818
5	DT	27.419	0.063643	0.170755

Interpretation By Experiment Block

1. The Full Strict Reference Is Now Operational

The most important campaign result is operational rather than purely numerical. The full strict exact-paper bank now completes end to end:

- all `10` recovered families fitted;
- all `20` harmonic targets evaluated per family;
- all `200` per-target ONNX artifacts exported;
- zero export failures under `strict` mode.

Interpretation:

- the exact-paper branch is no longer only prepared;
- the `HGBM` ONNX export blocker has been removed in the repository-owned support layer;
- the campaign now provides a stable exact-paper reference surface for future comparison and deployment-facing inspection.

2. The Diagnostic Continue Run Is Now A Safety Net, Not The Mainline

The full-bank diagnostic run now reaches the same winner metric as the strict reference:

- full diagnostic winner: `RF` at `18.369%`
- full strict winner: `RF` at `18.369%`

Interpretation:

- diagnostic mode remains useful for future isolation work;
- however, the branch no longer depends on diagnostic continuation to finish;
- promotion should therefore move to the strict full-bank run, not stay on the continue-mode path.

3. The `SVR` Branch Is Stable, But It Still Requires Surrogate Handling

The isolated `SVR` export diagnostic completed successfully:

- exported targets: `20`
- failed exports: `0`
- constant-surrogate exports: `5`

  Interpretation:
- the earlier degenerate- `SVR` converter crash is resolved operationally;
- the branch is still not a pure raw- `SVR` export path, because `5` targets

  remain constant predictors under the modern stack;
- the surrogate behavior is now a documented compatibility layer rather than a

  hidden failure.

4. The Non- `SVR` Branch Confirms The Remaining Bank Is Clean

The non- `SVR` reference run exported the remaining family bank independently:

- enabled families: `9`
- exported files: `180`
- failed exports: `0`

  Interpretation:
- the rest of the recovered family bank no longer depends on the `SVR`

  safeguard to remain operational;
- the branch can now be discussed as two stable pieces:
  - the special-case `SVR` branch with documented surrogate handling;
  - the non- `SVR` branch with clean strict export behavior.

Error And Family Findings

The winning strict full-bank run keeps the same best family ordering as the diagnostic full-bank run. The best exact-paper family in this repository branch is still:

- `RF`

  The highest-contributing winning-family target errors remain concentrated in the phase branch:
- `fft_y_Fw_filtered_phase_156` MAE `0.425070`
- `fft_y_Fw_filtered_phase_240` MAE `0.277042`
- `fft_y_Fw_filtered_phase_162` MAE `0.230457`
- `fft_y_Fw_filtered_phase_78` MAE `0.051585`
- `fft_y_Fw_filtered_phase_81` MAE `0.047979`

  Interpretation:
- the exact-paper branch remains dominated by phase-target difficulty rather

  than amplitude-target difficulty;
- the exact family-bank workflow is now stable enough that future work can

  focus on interpretation and parity analysis instead of export rescue.

ONNX Export Alignment Findings

The campaign succeeded operationally, but the path-level recovered-reference comparison remains intentionally unresolved:

- full-bank exported files: `200`
- recovered reference files: `201`
- matched relative paths: `0`
- missing relative paths against reference: `201`
- extra exported relative paths: `200`

  This does **not** mean the export failed. The mismatch is structural:
- the recovered paper bundle uses nested family-plus-kind paths such as

  `DT/ampl/DecisionTreeRegressor_ampl0.onnx`;
- the repository export currently writes flat family folders such as

  `DT/DecisionTreeRegressor_ampl0.onnx`;
- the recovered bundle also contains at least one duplicate-style filename

  variant such as `RF/ampl/RandomForestRegressor_ampl240 (1).onnx`. Interpretation:
- exportability is now stable;
- semantic path parity with the recovered release is still open;
- the next exact-paper refinement should normalize repository export layout if

  strict recovered-path mirroring is required.

Paper Alignment Impact

This campaign materially changes the exact-paper branch in three ways.

What Improved

- the exact-paper model-bank launcher is no longer only campaign-prepared;
- the full strict recovered family bank now executes end to end;
- the `SVR` and `HGBM` export blockers are both operationally contained.

What Remains Open

- the exact branch still uses component-level family metrics, not the harmonic

  reconstruction TE-curve benchmark used for `Target A`;
- recovered-path ONNX parity is still not normalized;
- no online compensation benchmark exists yet.

What The Campaign Established

- the exact-paper branch is now a stable structural reference bank;
- the harmonic-wise branch remains the current `Target A` offline benchmark;
- future paper-comparison reporting can now distinguish clearly between:
  - harmonic-wise TE-curve benchmark status;
  - exact family-bank export and parity status.

Main Conclusions

The campaign supports five conclusions.

1. The Exact-Paper Branch Is Now Campaign-Executable

The repository can now run the full exact-paper family bank through the canonical batch launcher without manual intervention.

2. The Correct Promoted Winner Is The Strict Full-Bank Reference

The diagnostic full-bank run and the strict reference reached the same winning metric, so promotion should prefer the strict exact-paper reference because it proves the branch no longer needs diagnostic continuation behavior.

3. The `HGBM` Export Issue Was Real And Is Now Resolved

The campaign would not have finished without the repository-side `HGBM` converter sanitation fix. That fix is now validated by the real four-run batch.

4. `SVR` Is Operationally Stable But Still Special-Case

The branch is exportable again, but only because the repository now documents and applies constant-surrogate handling for degenerate `SVR` targets.

5. Exact-Bank Success Does Not Yet Close `Target A`

This campaign stabilizes the exact-paper branch, but it does not replace the harmonic-wise offline TE-curve benchmark that still sits at `8.877%` and remains above the paper's `4.7%` threshold.

Recommended Next Actions

The next technically justified steps are

1. keep `exact_full_bank_strict_reference` as the canonical exact-paper run when discussing recovered-family-bank status;
2. preserve the `SVR` diagnostic branch for future surrogate-audit work rather than as the promoted reference;
3. decide whether the next exact-paper task should target recovered ONNX path mirroring, because the current export surface is operationally valid but not path-identical to the recovered release;
4. keep `Target A` tracking anchored to the harmonic-wise offline branch until the exact bank is connected to a TE-curve reconstruction benchmark;
5. defer any LAN-remote rerun unless future exact-bank iterations become materially heavier than the current local stabilization pass.

Artifact References

- Campaign root:
  `output/training_campaigns/exact_paper_model_bank_campaign_2026_04_10_17_04_41/`
- Campaign leaderboard:
  `output/training_campaigns/exact_paper_model_bank_campaign_2026_04_10_17_04_41/campaign_leaderboard.yaml`
- Campaign best run YAML:
  `output/training_campaigns/exact_paper_model_bank_campaign_2026_04_10_17_04_41/campaign_best_run.yaml`
- Campaign best run note:
  `output/training_campaigns/exact_paper_model_bank_campaign_2026_04_10_17_04_41/campaign_best_run.md`
- Winning strict validation summary:
  `output/validation_checks/paper_reimplementation_rcim_exact_model_bank/2026-04-10-19-12-14__exact_full_bank_strict_reference_campaign_run/validation_summary.yaml`
- Winning strict model bundle:
  `output/validation_checks/paper_reimplementation_rcim_exact_model_bank/2026-04-10-19-12-14__exact_full_bank_strict_reference_campaign_run/paper_family_model_bank.pkl`
- Diagnostic `SVR` validation summary:
  `output/validation_checks/paper_reimplementation_rcim_exact_model_bank/2026-04-10-19-12-57__exact_svr_export_diagnostic_campaign_run/validation_summary.yaml`
- Per-run launcher logs:
  `output/training_campaigns/exact_paper_model_bank_campaign_2026_04_10_17_04_41/logs/`